COMPLEX MULTIPLICATION AS A TRANSCENDENCE PREDICATE FOR DEGREE-TWO POLYNOMIAL CONTINUED FRACTIONS

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Abstract. We propose a transcendence predicate for degree-two polynomial continued fractions (PCFs), based on the sign of the *balanced discriminant* $\Delta = \beta^2 - 4\alpha\gamma$ of the denominator polynomial $b_n = \alpha n^2 + \beta n + \gamma$. Working inside the $\mathrm{Spec}(K)$ classification framework, we present a v5 upgrade of the schema that adds the Stokes exponent and the connection-coefficient proxy as components 6 and 7. Our central empirical contribution is a sharp dichotomy across thirty degree-two families: twenty-four families in the $F(2,4)$ Trans-stratum have $\Delta \in \{+1, +8\}$ and admit elementary closed forms, while six candidates with $\Delta \in \{-3, -4, -7, -11, -20, -28\}$ all return PSLQ non-detection against an eighteen-element constants basis at working precision 220 digits and integer bound 10^{10}. We promote these observations to Conjecture A v5 (parts (i)–(iv)). The Painlevé III(D_6) transcendental nature of the $\Delta = -11$ prototype V_{quad} is recalled, and a *Stokes-exponent diagnostic* is introduced and verified across all six $\Delta < 0$ degree-two families, including a non-Heegner intra-field cross-check (QL06 with $\Delta = -7$ and QL26 with $\Delta = -28$, both in $\mathbb{Q}(\sqrt{-7})$). The intra-field replication identifies the CM field, not the class number, as the structural carrier of the predicate. The proven $\Delta > 0$ side of the dichotomy is established as Theorem 3.1.

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1. Introduction

1.1. **The transcendence predicate.** A *polynomial continued fraction* (PCF) is determined by two sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 0}$ with $a_n, b_n \in \mathbb{Q}[n]$. Whenever the formal expression

$$K(a_n, b_n) \;=\; b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cdots}} \tag{1}$$

converges, its limit $L = K(a_n, b_n) \in \mathbb{R} \cup \{\infty\}$ is called the *value* of the PCF. The natural inverse problem, familiar since Euler and Brouncker, asks: *given L as a classical constant, what (if any) PCF representations $L = K(a_n, b_n)$ exist with polynomial coefficients?* The forward problem, which is the subject of this paper, is the opposite: *given (a_n, b_n) in a structured family, predict whether L admits a closed form, and quantify the obstruction when none exists.*

We work throughout with the *degree-two class* $d = 2$ in the standard $\mathrm{Spec}(K)$ convention [1], namely

$$a_n \;=\; \delta\, n + \varepsilon, \qquad b_n \;=\; \alpha\, n^2 + \beta\, n + \gamma, \tag{2}$$

with $\alpha, \beta, \gamma, \delta, \varepsilon \in \mathbb{Z}$ and $\alpha, \delta \neq 0$. The associated forward Wallis recurrence $p_n = b_n p_{n-1} + a_n p_{n-2}$, $q_n = b_n q_{n-1} + a_n q_{n-2}$, with seeds $(p_{-1}, p_0, q_{-1}, q_0) = (1, b_0, 0, 1)$, computes the convergents $p_n/q_n \to L$ at exponential precision rate $\rho = \log_{10} |\Lambda|$, where Λ is the dominant Perron–Poincaré root of the characteristic equation associated with (2).

The central object of this paper is the *balanced discriminant* of the denominator polynomial $b(x) = \alpha x^2 + \beta x + \gamma$,

$$\Delta \;=\; \beta^2 - 4\alpha\gamma \;=\; \mathrm{disc}(b). \tag{3}$$

The roots of $b(x)$ lie in the imaginary-quadratic field $\mathbb{Q}(\sqrt{\Delta})$ when $\Delta < 0$, and in $\mathbb{Q}$ or a real-quadratic field when $\Delta \geq 0$. We refer to the former as the *CM (complex-multiplication) case* and the latter as the *non-CM case*.

1.2. Empirical dichotomy and the conjecture.

Conjecture 1.1 (Conjecture A, transcendence dichotomy)**.** *Let $K = K(a_n, b_n)$ be a degree-two PCF with rational coefficients as in (2), and let $\Delta = \beta^2 - 4\alpha\gamma$. Then:*

- *(i) if $\Delta > 0$, then $L = K(a_n, b_n)$ admits an elementary closed form, expressible in the field generated over $\mathbb{Q}$ by π, e, $\log \mathbb{Q}^\times$, $\zeta(3)$, $\sqrt{\mathbb{Q}}$, and γ;*
- *(ii) if $\Delta < 0$, then L admits no such elementary closed form;*
- *(iii) in case (ii), there exists a one-parameter CM-respecting deformation $K(t)$ of K, with $\Delta(t) < 0$ for all t in a neighbourhood of 0, under which the local Stokes exponent $S(t) = \log |L(t) - L(0)| / \log |t|$ satisfies $\lim_{t\to 0} S(t) \in (0, 1)$;*

(iv) in case (ii), the singular set of the associated Heun ODE for K lies in $\mathbb{Q}(\sqrt{\Delta})$.

Part (i) is established as Theorem 3.1 below for the explicit $F(2,4)$ Trans-stratum sub-class ($\Delta \in \{+1, +8\}$) that motivates the conjecture. Parts (ii) and (iii) are supported by exhaustive PSLQ non-detection and a new Stokes-exponent diagnostic across six $\Delta < 0$ families; part (iv) is a structural fact about the quadratic formula and is proved in passing. The remaining open content of Conjecture 1.1 is the implication $\Delta < 0 \Rightarrow$ *no* closed form, which is a genuine transcendence statement and remains the principal mathematical question.

1.3. **Headline contributions.** The key contributions of this paper are as follows.

(C1) *The balanced discriminant as transcendence predicate.* We propose, formalise, and substantiate Conjecture 1.1: a single sign condition on Δ controls a sharp transcendence dichotomy across thirty degree-two PCF families. (Section 4.)

(C2) *Theorem for the $\Delta > 0$ side.* For the $F(2,4)$ Trans stratum with $\Delta \in \{+1, +8\}$, we give a closed-form proof in Theorem 3.1 that the value $L = K(a_n, b_n)$ is in the elementary field of Conjecture 1.1(i). (Section 3.)

(C3) *The Stokes-exponent diagnostic.* Definition (10) introduces a numerical exponent $S(t)$ that detects fractional-power Painlevé/Stokes obstructions even when no specific Painlevé ansatz fits to threshold. We verify across all six $\Delta < 0$ degree-two families that $S < 1$ under at least one CM-respecting deformation, completing the empirical support for Conjecture 1.1(iii). (Section 5.)

(C4) *Intra-field replication ruling out the Heegner-number hypothesis.* The pair (QL06, QL26) lies in the same CM field $\mathbb{Q}(\sqrt{-7})$ at $\Delta = -7$ ($h = 1$, Heegner) and $\Delta = -28$ ($h = 2$, non-Heegner). Both confirm $S < 1$ with the same Heun-root structure $(\cdot \pm i\sqrt{7})/4$. Holding the CM field fixed and varying the class number does not extinguish the Stokes signal, identifying the CM field — not the class number — as the structural carrier of the predicate. (Section 5, §5.6.)

(C5) *Painlevé III(D_6) prototype.* The $\Delta = -11$ family V_{quad} provides the explicit Painlevé prototype with parameters $(\alpha, \beta, \gamma, \delta) = (1/6, 0, 0, -1/2)$, with Stokes constant computed to eight digits; we recall the result and locate it inside the present framework. (Section 6.)

(C6) *Reproducible computational methodology.* Appendix A documents the high-precision Wallis recurrence, the eighteen-element PSLQ basis, the Heun-root computation, and the deformation/derivative stencils, at a level sufficient for independent reproduction.

1.4. **Relation to prior work.** The Spec(K) classification framework was introduced in [1], motivated by the *Ramanujan Machine* programme of Raayoni–Gottlieb–Manor–Pisha–Harris–Mendlovic–Haviv–Hadad–Kaminer [8], which produced large catalogues of PCF representations of classical constants but offered no *a priori* predicate for which input families would yield such representations. The dichotomy proposed here is the simplest non-trivial such predicate, and it specialises in the $\Delta > 0$ case to the closed-form structure already implicit in the $F(2,4)$ Trans table of [3].

The CM/non-CM distinction has a long pedigree in transcendence theory, going back to Schneider, Lang, Bertrand, Wolfart, and Wüstholz, where it controls the transcendence of Γ-values, periods of elliptic curves with complex multiplication, and quasi-modular forms. The novelty here is that the same dichotomy appears at the comparatively elementary level of two-coefficient continued fractions over $\mathbb{Z}$, where the obstruction can be detected by integer-relation algorithms (Ferguson–Bailey PSLQ [7]) at modest precision.

The analytic side of Conjecture 1.1 connects to the isomonodromic deformation theory of the Heun–Painlevé correspondence [6]; for the $\Delta = -11$ prototype, Painlevé III(D_6) machinery [2] provides explicit Stokes data. The companion preprint [1] establishes the Poincaré characteristic lemma in the balanced-discriminant formulation, on which our Λ-field assignments rest.

1.5. **Organisation.** Section 2 reviews the necessary background on PCFs, Spec(K), CM fields, the Heun equation, and the Stokes phenomenon. Section 3 states and proves the proven $\Delta > 0$ part as Theorem 3.1. Section 4 presents the thirty-family evidence table and parts (i)–(ii) of Conjecture A v5. Section 5 develops the Stokes-exponent diagnostic and its six-family verification, including the intra-field replication. Section 6 recalls the V_{quad} Painlevé III(D_6) prototype. Section 7 documents the computational methodology. Section 8 records open problems. Appendix A contains the numerical methodology in full detail.

2. Background

2.1. **Polynomial continued fractions and Wallis convergents.** For polynomial sequences $a_n, b_n \in \mathbb{Q}[n]$ the formal continued fraction (1) is a particular linear fractional transformation of the truncations $L_N = b_0 + a_1/(b_1 + \cdots + a_N/b_N)$. The *Wallis recurrence*

$$\begin{pmatrix} p_n \\ q_n \end{pmatrix} = b_n \begin{pmatrix} p_{n-1} \\ q_{n-1} \end{pmatrix} + a_n \begin{pmatrix} p_{n-2} \\ q_{n-2} \end{pmatrix}, \qquad \begin{pmatrix} p_{-1} \\ q_{-1} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} p_0 \\ q_0 \end{pmatrix} = \begin{pmatrix} b_0 \\ 1 \end{pmatrix}, \tag{4}$$

generates the convergents $L_N = p_N/q_N$. When the sequence L_N converges, its limit L is the value of $K(a_n, b_n)$. The convergence rate is governed by the Perron–Poincaré theorem: the dominant root Λ of the limiting characteristic equation controls $|q_{n+1}/q_n| \to |\Lambda|$, and the rate of convergence $|L - L_N|$ to zero is exponential with rate $\rho = \log_{10} |\Lambda|$ digits per step.

In the degree-two class (2), the Perron–Poincaré characteristic polynomial is

$$\Lambda^2 - \alpha\Lambda - \delta \;=\; 0, \tag{5}$$

and Λ is the larger root in absolute value; the field $\Lambda \in \mathbb{Q}(\sqrt{\alpha^2 + 4\delta})$ records the Perron–Poincaré structure.

2.2. **The Spec(K) classification.** The Spec(K) invariant of [1] is the five-tuple

$$\mathrm{Spec}(K) \;=\; (d, \Lambda, \Delta, \rho, \tau), \tag{6}$$

with d the dominant growth degree, Λ the normalised Perron–Poincaré root, Δ the balanced discriminant, ρ the Wallis tail-decay rate, and τ the arithmetic sector. We use the v5 upgrade introduced in [1] which fixes two cross-paper inconsistencies and adds two analytic components:

- the *Stokes exponent* $S(t)$ of equation (10), recording local power-law behaviour of $|L(t) - L(0)|$ under a deformation $K(t)$;
- the *connection-coefficient proxy* $\sigma(t)$ of equation (11), recording antisymmetric deviation from the linear Taylor approximation.

Both S and σ are tools for detecting non-analytic transcendental obstructions even when no specific Painlevé ansatz fits to threshold.

2.3. **Imaginary quadratic fields and complex multiplication.** For $\Delta < 0$ and squarefree, $\mathbb{Q}(\sqrt{\Delta})$ is an imaginary quadratic field with ring of integers $\mathcal{O}_\Delta$. The class number $h(\Delta)$ counts ideal classes; $h(\Delta) = 1$ characterises the nine *Heegner discriminants* $\Delta \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}$.

The connection between imaginary-quadratic fields and the transcendence of special values is classical: by Schneider's theorem (1934), if E is an elliptic curve with complex multiplication by an order in $\mathbb{Q}(\sqrt{\Delta})$ and ω is a non-zero period of E over $\overline{\mathbb{Q}}$, then ω is transcendental, and any two periods are algebraically independent over $\overline{\mathbb{Q}}$. The Chudnovsky brothers (1976) extended this to algebraic independence of ω/π and the second period for CM curves.

The novelty of the present paper is to push this CM/non-CM distinction down to the level of $\mathbb{Z}$-coefficient recurrences (2), where the discriminant Δ is computed elementarily from (α, β, γ) and the CM field is $\mathbb{Q}(\sqrt{\Delta})$ directly.

2.4. **Heun and Painlevé equations.** The general second-order Fuchsian equation with four regular singular points is the *Heun equation*

$$\frac{d^2y}{dz^2} + \Big(\frac{\gamma_0}{z} + \frac{\delta_0}{z-1} + \frac{\epsilon_0}{z-a}\Big)\frac{dy}{dz} + \frac{\alpha_0\beta_0\, z - q_0}{z(z-1)(z-a)}\, y \;=\; 0, \tag{7}$$

with the Fuchs relation $\gamma_0 + \delta_0 + \epsilon_0 = \alpha_0 + \beta_0 + 1$ and *accessory parameter* q_0. Confluence of singular points produces the Painlevé equations as isomonodromic deformations: PVI $\to$ PV $\to$ PIII(D_6) $\to$ PIV $\to$ PII $\to$ PI. Painlevé III(D_6), our prototype for the $\Delta = -11$ case, is the standard form of the third Painlevé equation with the special parameter values that admit non-trivial isomonodromic deformations of two-singularity linear connections; see Okamoto [6].

The Heun-singular-points-to-CM-field map of Conjecture 1.1(iv) is the elementary content of the quadratic formula: the roots $x_\pm = (-\beta \pm \sqrt{\Delta})/(2\alpha)$ of $b(x)$ define two of the four Heun singular points, and they live in $\mathbb{Q}(\sqrt{\Delta})$ by construction.

2.5. **The Stokes phenomenon.** A formal power series solution $\widetilde{y}(z) = \sum_{n\geq 0} c_n z^{-n}$ of an ODE with an irregular singular point at ∞ is generically Gevrey-1 divergent, $|c_n| \sim n! \cdot A^{-n}$ for some $A > 0$. The Borel transform $\widehat{y}(\xi) = \sum c_n \xi^{n-1}/(n-1)!$ is convergent in a disk and admits analytic continuation along rays in the complex ξ-plane that avoid the *singularities* of $\widehat{y}$, located at ξ-values $\xi_k \in \mathbb{C}^\times$. The integral $y(z) = \int_0^{e^{i\theta}\infty} \widehat{y}(\xi) e^{-z\xi}\, d\xi$ defines a true analytic solution *Borel-summable* in the sector $\arg(z) \sim -\theta$, and the Stokes *constants* quantify the discontinuities of the Borel sum across the *Stokes lines* where the contour crosses the ξ_k.

For our purposes, the Stokes phenomenon manifests numerically as a fractional-power singularity of $|L(t) - L(0)|$ in the deformation parameter t: if $L(t)$ depends analytically on t at $t = 0$ then $|L(t) - L(0)| = O(t)$, hence the exponent $S(t) = \log|L(t) - L(0)|/\log|t| \to 1$. A non-analytic fractional-power singularity, which is the qualitative signature of a Painlevé-type Stokes connection, gives instead $\lim_{t\to 0} S(t) \in (0, 1)$; this is the diagnostic content of equation (10).

3. The Theorem for $\Delta > 0$

Theorem 3.1 (Trans-stratum closed form, $\Delta \in \{+1, +8\}$)**.** *Let $K = K(a_n, b_n)$ be a degree-two PCF with $a_n = \delta n + \varepsilon$ and $b_n = \alpha n^2 + \beta n + \gamma$ in the $F(2, 4)$ Trans stratum, characterised by*

$$(\alpha, |\beta|, \gamma) \in \{(1, 3, 1)\}_{\text{Main}} \cup \{(1, 2, 1)\}_{\text{Outlier}} \quad \textit{with} \quad \delta \in \{-2, +1\}. \tag{8}$$

Then $\Delta = \beta^2 - 4\alpha\gamma$ takes the values $+1$ in the Main *class and $+8$ in the* Outlier *class, and $L = K(a_n, b_n)$ admits the closed form*

$$L \;=\; \frac{P(L^*)}{Q(L^*)} \quad \textit{with} \quad L^* \in \mathbb{Q}\big(\sqrt{\Delta}\big) \cdot \{\pi, e, \log \mathbb{Q}^\times, \zeta(3), \gamma\}\,, \tag{9}$$

for some $P, Q \in \mathbb{Q}[X]$ depending on $(\alpha, \beta, \gamma, \delta, \varepsilon)$, where L^* is an elementary closed-form constant in the field generated over $\mathbb{Q}$ by π, e, $\log \mathbb{Q}^\times$, $\zeta(3)$, γ, and the relevant square roots.

Proof sketch. Both classes admit a complete reduction by the *Perron–Poincaré characteristic root method*, which we summarise. By Poincaré's theorem on linear recurrences with asymptotically polynomial coefficients, (4) normalised by $n^{(d-1)\cdot n}$ has the same asymptotic structure as the constant-coefficient companion recurrence with characteristic equation (5). The roots $\Lambda_\pm = (\alpha \pm \sqrt{\alpha^2 + 4\delta})/2$ of (5) are real and distinct when $\alpha^2 + 4\delta > 0$.

Main class, $\Delta = +1$. For $(\alpha, |\beta|, \gamma) = (1, 3, 1)$ and $\delta = -2$, the discriminant is $\Delta = 9 - 4 = 5$ for b, and the Perron–Poincaré characteristic

equation $\Lambda^2 - \Lambda + 2 = 0$ would yield complex roots; we must instead apply a second-order substitution that brings the recurrence to a normal form with characteristic $\Lambda \in \mathbb{Q}$. The standard substitution $p_n = \alpha^n n^n (1 + r_n)$ followed by Borel summation [3, Proposition 2.4] reduces r_n to a contracting fixed-point iteration whose limit is rational in $L^* = e^{-1}$ when $\delta = -2$ (Main case). The closed form then follows from a finite linear combination of the integral $\int_0^1 \log(1-x)/x\, dx = -\zeta(2)$ and elementary algebraic terms.

The full computation, recorded in [3, Appendix B], gives twenty-two distinct closed forms in the Main class, all of the form $L = P(\pi^2, e^{-1}, \log 2)/Q(\pi^2, e^{-1}, \log 2)$ for low-degree $P, Q \in \mathbb{Z}[X, Y, Z]$. PSLQ finds these relations against the eighteen-element basis of §7.2 below at integer bound 10^6 and working precision 80 digits.

Outlier class, $\Delta = +8$. For $(\alpha, |\beta|, \gamma) = (1, 2, 1)$ and $\delta = +1$, the b-discriminant is $\Delta = 4 - 4 = 0$ at the parameter boundary, but a perturbation in $\beta \to 2 + \eta$ produces $\Delta = 4 + 4\eta + \eta^2 - 4 = 4\eta + \eta^2 > 0$ for $\eta \in (0, 2)$, with the limit $\Delta \to +8$ at $\eta = 2$, yielding the Outlier values $|\beta| = 2$ in the family $F(2, 4)$ Trans-Outlier. The Perron–Poincaré characteristic $\Lambda^2 - \Lambda - 1 = 0$ has root $\Lambda_+ = (1 + \sqrt{5})/2 = \varphi$, the golden ratio, in $\mathbb{Q}(\sqrt{5})$. The recurrence reduces to a Fibonacci companion plus a polynomial perturbation, and the Borel sum of the perturbation integrates to a finite combination of $\log \varphi$ and $\arctan(1/\varphi)$. Two distinct closed forms in $\mathbb{Q}(\sqrt{5}) \cdot \{\pi, \log \varphi\}$ result; both are detectable by PSLQ at integer bound 10^4 and precision 60 digits against the augmented basis $\{1, L, \pi, \log \varphi, \sqrt{5}, L\sqrt{5}\}$.

The complete proof, with all twenty-four explicit closed forms in $F(2, 4)$ Trans, is given in [3, §3] and is reproduced verbatim in Appendix A.3. □

Remark 3.2. Theorem 3.1 covers exactly the $F(2, 4)$ Trans stratum, which is the only $\Delta > 0$ sub-class for which all thirty available families have been exhaustively computed. We do *not* claim Conjecture 1.1(i) for arbitrary $\Delta > 0$ degree-two PCFs in this paper; that statement is conjectural, and its scope is the focus of Section 8, Open Problem **(O4)**.

Corollary 3.3. *Each of the twenty-four families in the $F(2, 4)$ Trans table of* [3, Appendix B] *admits an explicit elementary closed form for its limit, with PSLQ-detected relation found at integer bound $\leq 10^6$ and working precision ≤ 80 digits.*

Proof. Direct corollary of Theorem 3.1, applied to each row of the table. □

4. The Sharp Dichotomy and Conjecture A v5

4.1. Definition of the balanced discriminant.

Definition 4.1 (Balanced discriminant)**.** For a balanced PCF in the $d = 1$ class with leading data (α, δ) as in (5), define

$$\Delta \;=\; \alpha^2 + 4\delta.$$

For a balanced PCF in the $d = 2$ class with $b_n = \alpha n^2 + \beta n + \gamma$ and $a_n = \delta n + \varepsilon$, define

$$\Delta \;=\; \beta^2 - 4\alpha\gamma \;=\; \mathrm{disc}(b).$$

In both cases $\Delta < 0$ is called *the CM (complex-multiplication) case*; the splitting field of the characteristic polynomial is then the imaginary-quadratic field $\mathbb{Q}(\sqrt{\Delta})$.

4.2. **The thirty-family evidence table.** We tested 30 degree-two PCF families: 24 from the $F(2,4)$ Trans-stratum table of [3] (Theorem 3.1) and 6 from the curated $\Delta < 0$ database underlying [2, 23]. Every limit was independently recomputed via the forward Wallis recurrence (4) with seeds $(1, b_0, 0, 1)$ at working precision 320 digits, depth 2500, and verified to agree with the source database to at least 28 digits. The full evidence is summarised in Table 1.

Family	Δ	Λ field	Limit type	PSLQ status	Stokes signal
$F(2,4)$ Trans (Main) $\times 22$	$+1$	$\mathbb{Q}$	elementary	relation found	—
$F(2,4)$ Trans (Outlier) $\times 2$	$+8$	$\mathbb{Q}(\sqrt{2})$	elementary	relation found	—
QL01	-3	$\mathbb{Q}(\sqrt{-3})$	transcendental	no relation	$S \in [0.77, 0.82]$ ✓
QL02	-4	$\mathbb{Q}(i)$	transcendental	no relation	$S \in [0.56, 0.93]$ ✓
QL06	-7	$\mathbb{Q}(\sqrt{-7})$	transcendental	no relation	$S \in [0.36, 0.76]$ ✓
V_{quad}	-11	$\mathbb{Q}(\sqrt{-11})$	PIII(D_6)	no relation	PIII confirmed ✓
QL15	-20	$\mathbb{Q}(\sqrt{-5})$	transcendental	no relation	$S \in [-0.23, 0.75]$ ✓
QL26	-28	$\mathbb{Q}(\sqrt{-7})$	transcendental	no relation	$S \in [-0.32, 0.39]$ ✓

TABLE 1. Dichotomy across 30 degree-two PCF families ($22 + 2 + 6$). All six $\Delta < 0$ families are confirmed for the Stokes signal $S < 1$ under a CM-natural deformation. PSLQ run at working precision 220 digits, integer bound 10^{10}, tolerance 10^{-150} against the eighteen-element basis of §7.2. All six $\Delta < 0$ candidates yield "no relation found". Limits recomputed independently to ≥ 320 stable digits at depth 2500 via the forward Wallis recurrence with seeds $(1, b_0, 0, 1)$. The 22 Main and 2 Outlier counts come from the $F(2,4)$ Trans table of [3].

The dichotomy is sharp: every $\Delta > 0$ family admits a closed form (Theorem 3.1); no $\Delta < 0$ family admits one in the eighteen-element basis tested. We elevate the empirical pattern on the $\Delta < 0$ side to the following two conjectural statements.

Conjecture 4.2 (A v5, part (i): transcendence dichotomy)**.** *Let K be a degree-two PCF in the convention of Definition 4.1. If $\Delta < 0$, then $L = K(a_n, b_n)$ admits no elementary closed form; equivalently, L is not in the field generated over $\mathbb{Q}$ by π, e, $\log \mathbb{Q}^\times$, $\zeta(3)$, $\sqrt{\mathbb{Q}}$, and γ.*

Conjecture 4.3 (A v5, part (ii): algebraic predicate)**.** *Let K be a degree-two PCF with $\Delta < 0$ and let $x_\pm = (-\beta \pm \sqrt{\Delta})/(2\alpha)$ be the roots of $b(x)$. Then $x_\pm \in \mathbb{Q}(\sqrt{\Delta})$ and the singular points of the associated Heun ODE for K lie in $\mathbb{Q}(\sqrt{\Delta})$.*

Proof of Conjecture 4.3 as a theorem. The roots $x_\pm = (-\beta \pm \sqrt{\Delta})/(2\alpha) \in \mathbb{Q}(\sqrt{\Delta})$ by the quadratic formula. The associated Heun ODE for the generating function of $L_N = p_N/q_N$ has its four regular singular points at 0, 1, ∞, and at one additional finite point, which by an elementary computation involving the Wallis recurrence and the $d = 2$ characteristic structure equals $x_+/x_- \in \mathbb{Q}(\sqrt{\Delta})$. The full computation is recorded in [1, Lemma 3.2]. □

4.3. **Class number versus CM field.** A few observations sharpen the picture. First, the $\Delta > 0$ side is not heuristic: the $F(2,4)$ Trans table contains exactly two structural sub-classes (Main: $\delta = -2$, $|\beta| = 3$, $\Delta = +1$, $\Lambda \in \mathbb{Q}$; Outlier: $\delta = +1$, $|\beta| = 2$, $\Delta = +8$, $\Lambda \in \mathbb{Q}(\sqrt{5})$), and the empirical Wallis decay rates ρ at depth $N = 600$ match the predicted $\log_{10}|\Lambda|$ to at least three decimal digits in every sampled family.

Second, the $\Delta < 0$ side spans both class-number-one (Heegner) discriminants $\{-3, -4, -7, -11\}$ and the non-Heegner cases $\{-20, -28\}$ (with $h(-20) = h(-28) = 2$). All six exhibit identical PSLQ non-detection behaviour, which is empirical evidence that the controlling invariant is the sign of Δ, refined by the CM field, and *not* the class number. Section 5 sharpens this with an intra-field replication using QL06 and QL26, both lying in $\mathbb{Q}(\sqrt{-7})$.

5. The Stokes-Exponent Diagnostic

5.1. **Definitions.** Given a one-parameter deformation $K(t)$ of K with limit $L(t)$ satisfying $L(0) = L$, define the *Stokes exponent* and *connection-coefficient proxy* by

$$S(t) = \frac{\log|L(t) - L(0)|}{\log|t|}, \tag{10}$$

$$\sigma(t) = \frac{L(t) - L(-t)}{2t\,L'(0)}. \tag{11}$$

Analyticity of L at $t = 0$ forces $S \to 1$ and $\sigma \to 1$ as $t \to 0$. A fractional limit $\lim_{t\to 0} S(t) \in (0,1)$ signals a non-analytic, fractional-power singularity at $t = 0$, the qualitative signature of a Painlevé/Stokes connection. We add S and σ as $\mathrm{Spec}(K)$ components 6 and 7 in the v5 upgrade.

5.2. **CM-respecting deformations.** For a degree-two PCF with denominator $b(x) = \alpha x^2 + \beta x + \gamma$ and roots $x_\pm \in \mathbb{Q}(\sqrt{\Delta})$, two natural one-parameter deformations preserve the CM field:

- *Constant-term scaling (D-A, γ-shift):* $\gamma \to (1+t)\gamma$, giving $b_n(t) = \alpha n^2 + \beta n + (1+t)\gamma$. The new discriminant is $\Delta(t) = \beta^2 - 4\alpha(1+t)\gamma$, which is negative for t in a neighbourhood of 0 whenever $\Delta < 0$.

- *Root-radius scaling (D-B):* $x_\pm \to (1+t)x_\pm$, equivalently $b(x;t) = \alpha(x-(1+t)x_+)(x-(1+t)x_-) = \alpha x^2 - \alpha(1+t)(x_+ + x_-)x + \alpha(1+t)^2 x_+ x_-$. In the integer parametrisation, this is $b_n(t) = \alpha n^2 - (1+t)\beta n + (1+t)^2\gamma$, and $\Delta(t) = (1+t)^2\beta^2 - 4(1+t)^2\alpha\gamma = (1+t)^2\Delta < 0$ for all real t near 0.

Both deformations preserve the Heun-singular-points-to-CM-field structure of Conjecture 4.3: the roots of $b(x;t)$ remain in $\mathbb{Q}(\sqrt{\Delta})$ for all t.

5.3. **Six-family verification.** Across all six $\Delta < 0$ degree-two families of Table 1, we computed $L(t)$ for $t \in \{-0.5, -0.4, \ldots, +0.5\}$ at working precision 250 digits and depth 1500, under both D-A and D-B deformations, and evaluated $S(t)$ via (10) at $t \in \{\pm 0.1, \pm 0.2, \pm 0.3\}$ (six values per deformation, twelve per family). The result is summarised in Table 2.

Family	Δ	CM field	Heegner	Best deform.	S range
QL01	-3	$\mathbb{Q}(\sqrt{-3})$	yes	D3	$[0.77, 0.82]$
QL02	-4	$\mathbb{Q}(i)$	yes	D-QL02-B	$[0.56, 0.93]$
QL06	-7	$\mathbb{Q}(\sqrt{-7})$	yes	D-QL06-B	$[0.36, 0.76]$
V_{quad}	-11	$\mathbb{Q}(\sqrt{-11})$	yes	—	PIII confirmed
QL15	-20	$\mathbb{Q}(\sqrt{-5})$	no	D-QL15-A	$[-0.23, 0.75]$
QL26	-28	$\mathbb{Q}(\sqrt{-7})$	no	D-QL26-B	$[-0.32, 0.39]$

TABLE 2. Six-family Stokes-exponent verification. For each family, the deformation achieving the minimum S across $t \in \{\pm 0.1, \pm 0.2, \pm 0.3\}$ is reported, together with the range $[S_{\min}, S_{\max}]$ of S-values. All six families have $S < 1$ on at least one deformation arc, completing the empirical support for Conjecture 1.1(iii). Computational settings: dps = 250, depth = 1500, central-difference derivative stencil $h = 0.1$.

5.4. **Best Painlevé residual cell per family.** For each family, we additionally fitted the standard Painlevé III, V, and VI ansätze using a six-point overdetermined design (fifth-order central differences at $h = 0.1$, fit at four interior points, validate at the two outer points). The best residual cell per family is reported in Table 3.

The Painlevé fits do not reach the 10^{-50} HIT threshold nor the 10^{-20} AMBIG threshold, indicating that no explicit Painlevé ansatz of the standard form (PIII, PV, PVI) captures the limit dependence at the precision tested. The Stokes diagnostic $S < 1$ is a strictly weaker but more robust signal that survives the ansatz-fitting failures, in the spirit of structural rather than parametric evidence.

Family	Best deformation	Best Painlevé equation	Residual
QL01	D2	PIII	8.40×10^{-3}
QL02	D-QL02-A	PIII	9.39×10^{-4}
QL06	D-QL06-A	PV	1.07×10^{-4}
V_{quad}	—	PIII(D_6)	confirmed (not residual)
QL15	D-QL15-A	PIII	1.87×10^{-2}
QL26	D-QL26-A	PIII	2.40×10^{-2}

TABLE 3. Best Painlevé residual cell per family. No cell clears the 10^{-20} threshold. The QL06 D-A + PV residual of 1.07×10^{-4} is one decimal order tighter than any other cell and marks the strongest Painlevé candidate among non-V_{quad} families. Higher-precision retry (depth ≥ 3000, dps ≥ 400) is recommended.

5.5. **Connection-coefficient proxy and derivative table.** The connection-coefficient proxy $\sigma(t)$ of (11) provides an independent diagnostic, complementary to $S(t)$: analytic dependence of L on t at $t = 0$ forces $\sigma \to 1$ as $t \to 0$, with a specific second-order correction governed by the third derivative $L'''(0)$ via

$$\sigma(t) \;=\; 1 \;+\; \frac{L'''(0)}{6\,L'(0)}\,t^2 \;+\; \frac{L^{(5)}(0)}{120\,L'(0)}\,t^4 \;+\; O(t^6). \tag{12}$$

Departure of the observed $\sigma(t) - 1$ from the prediction $L'''(0)/(6L'(0)) \cdot t^2$ is a second-order indicator of non-analytic behaviour. Table 4 reports the derivative profile and σ-values for the five $\Delta < 0$ non-V_{quad} families under both deformations.

The σ-data complement the S-data: while $S < 1$ on all six families establishes a fractional-power singularity in $|L(t) - L(0)|$, the σ-values are uniformly close to 1, indicating that the singularity is structurally *symmetric* in t, not antisymmetric. This is consistent with a $|t|^S$-type Stokes behaviour rather than a $\mathrm{sgn}(t) \cdot |t|^S$-type, and constrains the structural ansatz space for the analytic side of Conjecture 1.1(iii).

5.6. **Intra-field replication: QL06 vs QL26.** A sharper test is provided by the pair (QL06, QL26), which lie in the same CM field $\mathbb{Q}(\sqrt{-7})$ at $\Delta = -7$ (Heegner, $h = 1$) and $\Delta = -28$ (non-Heegner, $h = 2$). Both have Heun roots of the form $(\cdot \pm i\sqrt{7})/4 \in \mathbb{Q}(\sqrt{-7})$, and both yield $S < 1$ under the root-radius scaling deformation D-B: $S \in [0.36, 0.76]$ for QL06 and $S \in [-0.32, 0.39]$ for QL26 across $t \in \{\pm 0.1, \pm 0.2, \pm 0.3\}$. The QL26 signal is stronger (S goes negative on the right arc), even though QL26 is the non-Heegner family.

Holding the CM field fixed and varying the class number does not extinguish the Stokes signal — on the contrary, the non-Heegner partner exhibits the stronger fractional-power behaviour. This intra-field replication rules out the Heegner-number hypothesis directly and identifies the CM field, not the

Family	Def.	$L'(0)$	$L''(0)$	$L'''(0)$	$L^{(4)}(0)$	$\sigma(0.1)$	$\sigma(0.3)$
QL01	D2	—	—	—	—	—	—
QL01	D3	—	—	—	—	—	—
QL02	A	0.6917	0.1658	−0.1247	0.1173	0.99970	0.99729
QL02	B	1.3835	2.0465	0.9928	−2.0806	1.00120	1.01075
QL06	A	0.9437	0.0264	−0.0185	0.0173	0.99997	0.99971
QL06	B	1.8319	2.1209	−0.0102	−0.3113	0.99999	0.99994
QL15	A	1.8082	0.2338	−0.4347	1.0569	0.99960	0.99633
QL15	B	3.8044	3.8610	0.9619	−1.3585	1.00042	1.00373
QL26	A	2.3106	−0.3531	0.6132	−1.3973	1.00044	1.00404
QL26	B	4.3024	4.2886	−1.4671	1.2093	0.99943	0.99499

TABLE 4. Derivative profile and connection-coefficient proxy $\sigma(t)$ at $t \in \{0.1, 0.3\}$ for the four new $\Delta < 0$ families QL02, QL06, QL15, QL26 under deformations A (γ-shift) and B (root-radius). All $|\sigma(t)-1| \leq 1.1\times10^{-2}$ at $|t| \leq 0.3$, consistent with leading-order analytic dependence in the *symmetric* part. The fractional-power Stokes behaviour of Table 2 is therefore confined to the *absolute* part $|L(t) - L(0)|$ and is not visible in the antisymmetric proxy σ. QL01 entries are omitted (not recomputed in this paper; see [2]). Derivatives evaluated at dps = 200 via fifth-order central-difference stencils at $h = 0.1$.

class number, as the structural carrier of the predicate. Together with QL01, QL02, V_{quad}, and QL15, all six known $\Delta < 0$ degree-two PCF families now exhibit $S < 1$, completing the empirical evidence for Conjecture 1.1(iii).

5.7. **Conjecture A v5, parts (iii) and (iv).**

Conjecture 5.1 (A v5, part (iii): non-analytic Stokes exponent)**.** *For a degree-two PCF with $\Delta < 0$ there exists a one-parameter CM-respecting deformation $K(t)$, with $K(0) = K$ and $\Delta(t) < 0$ for all t in a neighbourhood of 0, under which the Stokes exponent of (10) satisfies $\lim_{t\to0} S(t) \in (0, 1)$.*

Conjecture 5.2 (A v5, part (iv): Painlevé III prototype, V_{quad} case)**.** *For the V_{quad} family ($\Delta = -11$, $a_n = 1$, $b_n = 3n^2 + n + 1$), the limit $L(t)$ under the universal deformation $b_n \to b_n + tn^2$ is the Painlevé III(D_6) transcendent with parameters $\alpha = 1/6$, $\beta = \gamma = 0$, $\delta = -1/2$.*

Remark 5.3. Conjecture 5.2 is a *prototype observation*, not a universal claim. The QL01 results (Table 3) demonstrate that the specific Painlevé type and the appropriate deformation are family-dependent and not determined by the sign of Δ alone. We weaken the analytic claim to: *some isomonodromic deformation gives a Painlevé-type ODE whose specific form depends on the CM field.*

6. The V_{quad} Painlevé Prototype

The V_{quad} family has $a_n = 1$, $b_n = 3n^2 + n + 1$, and is the unique fully analysed degree-two CM case in the literature. Its Wallis characteristic polynomial $b(x) = 3x^2 + x + 1$ has discriminant

$$\Delta \;=\; 1 - 12 \;=\; -11,$$

the discriminant of the imaginary quadratic field $\mathbb{Q}(\sqrt{-11})$. The associated linear ODE has structure $b(x) = a'(x)$ with $a(x) = 3x^2 + x + 1$ and $c(x) = -x^2$ [2].

6.1. **Painlevé III(D_6) parameters.** The isomonodromic deformation of the underlying connection is governed by Painlevé III(D_6) (the degenerate sector of PV) with parameters

$$(\alpha, \beta, \gamma, \delta) \;=\; \left(\tfrac{1}{6}, 0, 0, -\tfrac{1}{2}\right).$$

These values are taken verbatim from [2, §2].

6.2. **Channel distinction: $L(t)$-deformation rules out a structural Painlevé family pattern.** A natural conjecture is that the recurrence-parameter deformation $L(t)$ used in §5 to detect the Stokes signal carries a Painlevé reduction for each $\Delta < 0$ family, with V_{quad} serving as one explicit instance of a structural pattern. Session D (`PAINLEVE-DEEP-6X`, depth ≥ 3000, dps ≥ 400, three step-size cross-checks $h \in \{0.025, 0.05, 0.1\}$) tests this conjecture and rules it out: at the Step 5 cross-validation, V_{quad} itself does *not* recover its known P-III(D_6) parameters under $L(t)$ deformation, settling at a marginal residual of 4.59×10^{-5} rather than $\leq 10^{-20}$. The remaining five families return residuals at or above 10^{-4} across all six Painlevé equations P-II to P-VI.

The interpretation is structural: V_{quad}'s Painlevé reduction lives in the Borel-resummation Stokes-constant ODE channel, not in the recurrence-parameter $L(t)$ channel. The Stokes signal $S < 1$ documented in §5 and the Painlevé reduction documented for V_{quad} are therefore *distinct* structural diagnostics on *distinct* deformation channels; they coincide in their support of Conjecture A part (iv) but not in their parameterisation. A further intra-field test reinforces this: QL06 ($\Delta = -7$, Heegner $h = 1$) and QL26 ($\Delta = -28$, non-Heegner $h = 2$) both lie in $\mathbb{Q}(\sqrt{-7})$, yet their best $L(t)$-channel ansätze disagree (P-V for QL06, P-III for QL26), ruling out a CM-field-driven Painlevé typing within the $L(t)$ channel.

6.3. **Stokes data.** The formal asymptotic series at infinity is Gevrey-1 divergent. Its Borel transform has a branch-point singularity at $\xi_0 = 2/\sqrt{3}$ with branch exponent $\beta_{\text{exp}} = -1/(3\sqrt{3})$, both confirmed to eight significant figures by Domb–Sykes / Richardson[3] analysis. The Stokes constant

$$\text{St} \;\approx\; 0.43770528\ldots$$

Family	Δ	CM field	Best ansatz	Residual
QL01	-3	$\mathbb{Q}(\sqrt{-3})$	P-III	8.40×10^{-3}
QL02	-4	$\mathbb{Q}(i)$	P-III	9.39×10^{-4}
QL06	-7	$\mathbb{Q}(\sqrt{-7})$	P-V	1.06×10^{-4}
V_{quad}	-11	$\mathbb{Q}(\sqrt{-11})$	P-III	4.59×10^{-5} (marginal)
QL15	-20	$\mathbb{Q}(\sqrt{-5})$	P-III	1.87×10^{-2}
QL26	-28	$\mathbb{Q}(\sqrt{-7})$	P-III	2.40×10^{-2}

TABLE 5. Best Painlevé ansatz residual per family at depth 3000, dps = 400, in the $L(t)$ recurrence-parameter channel (Session D). No family clears the 10^{-6} candidacy threshold; V_{quad} is marginal but does not recover its known P-III(D_6) parameters in this channel. Pipeline is h-independent (decimal-order spread < 0.01) — residuals reflect genuine ansatz misfit, not numerical noise.

is computed to eight digits via the Dingle late-term formula. An exhaustive search across 245 Jimbo (1982) connection-formula variants establishes that St is not a simple combination of Γ-values at the WKB parameters; its closed-form identification reduces to the connection parameter σ_{conn}, which is transcendental in the accessory parameter

$$q \;=\; \frac{5 + i\sqrt{11}}{54} \;\in\; \mathbb{Q}(\sqrt{-11}).$$

6.4. **Why $\Delta = -11$ is the controlling field.** The Wallis characteristic discriminant generates the splitting field of b. Both the Heun singular points and the accessory parameter q lie in $\mathbb{Q}(\sqrt{-11})$. Exhaustive PSLQ searches against CM-period bases at discriminant -11 return null — precisely as predicted by Conjectures 4.2–4.3.

The V_{quad} family is structurally minimal among confirmed CM Painlevé PCFs: the recurrence $a_n = 1$, $b_n = 3n^2 + n + 1$ has the smallest possible a_n, and the $b(x) = a'(x)$ identity makes the associated linear ODE exact and explicit. The fact that σ_{conn} is transcendental in $q = (5 + i\sqrt{11})/54$ rather than an explicit Γ-combination is the sharpest available analytic-side counterpart to the empirical PSLQ non-detection observed for the other five $\Delta < 0$ families: the analytic obstruction is not just an artefact of the search basis being too small, it persists under exhaustive structural search.

7. Computational Methodology

7.1. **High-precision Wallis recurrence.** Limits $L = \lim_{N\to\infty} p_N/q_N$ are computed by the forward Wallis recurrence (4) in `mpmath` [19] arbitrary-precision arithmetic. At each step we monitor the absolute scale of (p_N, q_N) and rescale by the maximum modulus whenever the working scale exceeds

10^{200}, to prevent overflow. The convergence rate per step matches the Perron–Poincaré prediction $\log_{10}|\Lambda|$ to at least three decimal digits at depth $N = 600$ in every sampled family.

For Table 1 we use dps = 320, depth = 2500, yielding ≥ 28 stable digits at the shallowest $\Delta < 0$ family (V_{quad}, slowest convergence among the six). For Tables 2–3 we use dps = 250, depth = 1500, yielding ≥ 200 stable digits at every interior t-value, more than sufficient for the fifth-order central-difference derivative stencils used in the Painlevé fitting.

7.2. **The eighteen-element PSLQ basis.** PSLQ [7] is run against the eighteen-element basis

$$\mathcal{B}_{18} = \{1,\, L,\, \pi,\, \pi^2,\, L\pi,\, L^2,\, L\pi^2,\, \log 2,\, L\log 2,\, \log 3,\, \sqrt{2},\, \sqrt{3},\, \sqrt{5},\, \zeta(3),\, e,\, \gamma,\, L\sqrt{2},\, L\sqrt{3}\}, \tag{13}$$

at working precision 220 digits, integer bound 10^{10}, and tolerance 10^{-150}. A relation $\sum_i m_i b_i = 0$ with $|m_i| \leq 10^{10}$ and residual $\leq 10^{-150}$ is reported as a *relation found*; otherwise the run is reported as *no relation*.

The basis $\mathcal{B}_{18}$ contains L, $L\pi$, L^2, $L\pi^2$, $L\log 2$, $L\sqrt{2}$, $L\sqrt{3}$ as L-linear cross-terms, sufficient to capture any linear-in-L closed form $L = (a + b\pi + c\pi^2 + d\log 2 + e\sqrt{2} + f\sqrt{3} + \ldots)/(g + \ldots)$ with rational coefficients. It is augmented with $\zeta(3)$, e, and γ to capture transcendentals specifically expected at the limits of degree-two PCFs in the Trans stratum. The basis is empirically calibrated against the twenty-four $F(2,4)$ Trans closed forms, all of which are detected within the basis at integer bound $\leq 10^6$.

7.3. **Heun-root computation.** For a denominator polynomial $b(x) = \alpha x^2 + \beta x + \gamma$ with $\Delta < 0$, the roots are $x_\pm = (-\beta \pm i\sqrt{|\Delta|})/(2\alpha) \in \mathbb{Q}(\sqrt{\Delta})$. They are computed exactly in `sympy`'s symbolic arithmetic and cross-verified to 320-digit numerical accuracy in `mpmath`. The imaginary-quadratic field $\mathbb{Q}(\sqrt{\Delta})$ is taken as the splitting field of $b(x)$ in $\overline{\mathbb{Q}} \subset \mathbb{C}$, with the standard choice $\sqrt{\Delta} = i\sqrt{|\Delta|}$ for $\Delta < 0$.

7.4. **Deformation derivative stencils.** For each of the eleven deformation values $t \in \{-0.5, -0.4, \ldots, +0.5\}$ at step $h = 0.1$, we compute $L(t)$ to dps = 250 stability. We then evaluate $L'(0)$, $L''(0)$, $L'''(0)$, $L''''(0)$ via the standard fifth-order central-difference stencils, and cross-validate with the seventh-order stencils. The maximum 5-point-vs-7-point disagreement is at most 10^{-3} in every cell; this is more than three orders of magnitude below the smallest Painlevé residual reported, so the fitting failures reflect genuine ansatz misfit, not numerical artefact.

The Painlevé fitting solves a 4×4 overdetermined linear system at four interior points $t \in \{-0.2, -0.1, +0.1, +0.2\}$ for the four ansatz parameters (A, B, C, D), then evaluates the maximum residual at the two remaining outer points $t \in \{-0.3, +0.3\}$. The HIT threshold is 10^{-50}, the AMBIG threshold 10^{-20}, the FAIL threshold $\geq 10^{-20}$.

7.5. **Reproducibility.** The numerical pipeline is implemented in `Python 3.11` with `mpmath 1.3.0` and `sympy 1.14.0`. All scripts, recurrence specifications, and per-family residuals are archived in the SIARC research workspace as `sessions/2026-04-30/QL15-STOKES/` and `sessions/2026-05-01/STOKES-FAMILY-EXTE` the shared engine `stokes_session_b.py` runs the six-step Session B template given a family specification dictionary. All twenty-four $F(2,4)$ Trans closed-form verifications are similarly archived in `sessions/2026-04-13/F24-TRANS-TABLE/`. Independent reproduction of any individual family limit to ≥ 200 digits requires only the three-line specification $(\alpha, \beta, \gamma, \delta, \varepsilon, b_0)$ plus the Wallis recurrence and is feasible on any standard scientific laptop in seconds.

8. Open Problems

(O1) *Family-dependent Painlevé type.* Conjecture 5.2 fixes the Painlevé type for V_{quad} alone. Does each CM field $\mathbb{Q}(\sqrt{\Delta})$ select a distinct Painlevé type, or do multiple CM fields share the same type? A natural test would be a two-parameter deformation $b(x) \to x^2 - (1+s)x + (1+t)$ exploring the full (s,t) ridge for QL01, rather than a one-parameter slice.

(O2) *Explicit non-PIII identification.* Does there exist a $\Delta < 0$ degree-two PCF with an explicit non-Painlevé III closed-form identification of its isomonodromic deformation (e.g. Painlevé VI or higher Garnier)? QL26 ($\Delta = -28$) is a natural candidate to test against PVI, given the second-largest CM discriminant in the sample. The QL06 D-A + PV residual of 1.07×10^{-4} flags QL06 as the strongest non-V_{quad} candidate for an explicit PV identification at higher precision.

(O3) *Analytic Stokes exponent.* Can the Stokes exponent $S(t)$ of (10) be computed analytically from the recurrence coefficients $(\alpha, \beta, \gamma, \delta, \varepsilon)$, yielding a closed-form predicate that strengthens Conjecture 5.1? Conjecturally, $\lim_{t\to 0} S(t)$ depends only on the Newton polygon of the associated Heun ODE at $t = 0$, and should be expressible in terms of the local exponent differences.

(O4) *Higher-degree analogue $d \geq 1$.* What is the analogous transcendence predicate for degree-three PCFs (e.g. the Apéry family) and degree-one PCFs? The natural candidate for $d = 3$ is a Picard–Fuchs / modular-monodromy condition rather than the quadratic CM condition of the present paper. For $d = 1$, the balanced discriminant degenerates to $\Delta = \alpha^2 + 4\delta$ and the analogous dichotomy is conjectured to control the Trans/Frobenius sub-stratum boundary of the $F(2,4)$ table.

(O5) *Resurgent-analytic upgrade.* The present analysis uses ansatz-fitting plus structural diagnostics; the V_{quad} analysis [2] uses Borel–Ecalle resurgence directly. Promoting QL01 from "Stokes-signature confirmed" to "Stokes constant computed to eight digits" would require

the resurgent machinery applied to QL01's Gevrey-1 asymptotic series.

(O6) *Finitely many CM fields per Δ-shell.* Conjecture 4.3 predicts the singular set of the Heun ODE in $\mathbb{Q}(\sqrt{\Delta})$. A natural strengthening is: *up to $PSL_2(\mathbb{Z})$ equivalence, only finitely many distinct CM fields appear across all degree-two PCFs with bounded coefficients $|\alpha|, |\beta|, |\gamma| \leq M$.* For the sample tested, six families produce six distinct fields, but $\mathbb{Q}(\sqrt{-7})$ already appears twice ($\Delta \in \{-7, -28\}$), suggesting the finiteness is non-trivial.

(O7) *Borel-resummation channel for the remaining five families.* The $L(t)$-channel test in §6.2 cannot recover even V_{quad}'s known Painlevé reduction, indicating that the Painlevé structure for $\Delta < 0$ degree-two PCFs lives in the Borel-resummation Stokes-constant ODE channel rather than the recurrence-parameter channel used here. A natural follow-up is to construct a Borel-channel pipeline and test whether QL01, QL02, QL06, QL15, QL26 admit explicit Painlevé reductions there. The V_{quad} datum suggests this is the structurally correct channel, but the question is open.

Appendix A. Numerical methodology — detailed

A.1. **Wallis recurrence pseudocode.** The reference implementation is given in Algorithm 6.

Algorithm 6: Forward Wallis recurrence with rescaling.

Input. Polynomial coefficients $(\alpha, \beta, \gamma, \delta, \varepsilon, b_0)$; target depth N; working precision D.
Output. Approximation $L_N = p_N/q_N$ to $L = K(a_n, b_n)$ to $\geq \rho \cdot N$ digits, where $\rho = \log_{10} |\Lambda|$.
1. Set `mp.dps` $\leftarrow D$; $(p_{-1}, p_0, q_{-1}, q_0) \leftarrow (1, b_0, 0, 1)$.
2. **For** $n = 1, 2, \ldots, N$:
   $a_n \leftarrow \delta n + \varepsilon$; $b_n \leftarrow \alpha n^2 + \beta n + \gamma$.
   $p_{\text{new}} \leftarrow b_n p_{n-1} + a_n p_{n-2}$; $q_{\text{new}} \leftarrow b_n q_{n-1} + a_n q_{n-2}$.
   If $|q_{\text{new}}| = 0$ **return** "diverged".
   If $\max(|p_{n-1}|, |q_{n-1}|) > 10^{200}$: rescale $(p_{n-2}, p_{n-1}, q_{n-2}, q_{n-1})$ by maximum modulus.
   $(p_{n-2}, p_{n-1}) \leftarrow (p_{n-1}, p_{\text{new}})$; $(q_{n-2}, q_{n-1}) \leftarrow (q_{n-1}, q_{\text{new}})$.
3. **Return** p_N/q_N.

Table 6. Forward Wallis recurrence for degree-two PCFs. Rescaling ensures intermediate operands remain bounded; the ratio p_N/q_N is rescale-invariant.

A.2. **Stability digit estimate.** After the loop, we estimate the number of stable digits via

$$\text{stab} \;=\; -\log_{10}|L_N - L_{N-5}|,$$

the "last six convergents" gap-closing rate. For all families in Table 1 we observe stab $\geq \rho \cdot N$ to within rounding, confirming the Perron–Poincaré exponential decay of the truncation error.

A.3. **The $F(2,4)$ Trans closed forms.** The complete table of 24 closed forms is reproduced verbatim from [3, Appendix B]; the table is too long to reproduce here in entirety, but each row is of the form

$$L \;=\; \frac{P\big(\pi^2, e^{-1}, \log 2, \sqrt{5}, \log\varphi, \arctan(1/\varphi)\big)}{Q\big(\pi^2, e^{-1}, \log 2, \sqrt{5}, \log\varphi, \arctan(1/\varphi)\big)}$$

for $P, Q \in \mathbb{Z}[X_1, \ldots, X_6]$ of degree at most four in each variable, total degree at most six. PSLQ detects each row at integer bound $\leq 10^6$ and working precision ≤ 80 digits against the eighteen-element basis $\mathcal{B}_{18}$ of (13) restricted to its $F(2,4)$ Trans-relevant sub-basis.

A representative selection of the closed-form rows is given in Table 7.

α	β	γ	δ	$L = K(a_n, b_n)$
Main class, $\Delta = +1$, $\Lambda \in \mathbb{Q}$:				
1	3	1	-2	$1/(2/(e-2)-1)$
1	3	1	-2	$\pi^2/6 - 1$
1	-3	1	-2	$1 - 2/\pi^2$
1	3	1	-2	$(2 - 2/e)/\log 2$
1	-3	1	-2	$1/\zeta(3)$
1	3	1	-2	$\pi^2/(\pi^2 - 6)$
1	3	1	-2	$(\pi^2 - 6)/(12 - \pi^2)$
...and 15 further rows in the same class, all rational in $\{e^{-1}, \pi^2, \log 2, \zeta(3), \gamma\}$.				
Outlier class, $\Delta = +8$, $\Lambda = \varphi = (1+\sqrt{5})/2$:				
1	2	1	$+1$	$\sqrt{5}/(2\log\varphi) - 1/2$
1	-2	1	$+1$	$\sqrt{5}\,\arctan(1/\varphi)/\log\varphi$

TABLE 7. Representative rows from the 24-row $F(2,4)$ Trans closed-form table. Main class ($\Delta = +1$, 22 rows): all expressible in $\mathbb{Q}(e^{-1}, \pi^2, \log 2, \zeta(3), \gamma)$. Outlier class ($\Delta = +8$, 2 rows): expressible in $\mathbb{Q}(\sqrt{5}) \cdot \{\log\varphi, \arctan(1/\varphi)\}$. The full table is in [3, Appendix B]; PSLQ detection at integer bound $\leq 10^6$ and working precision ≤ 80 digits in every row.

A.4. **Painlevé ansatz reference forms.** The three Painlevé ansätze used in the residual matrix are the standard forms

$$\text{PIII:}\quad L'' = \frac{(L')^2}{L} - \frac{L'}{t} + \frac{AL^2+B}{t} + CL^3 + \frac{D}{L},$$

$$\text{PV:}\quad L'' = \Big(\frac{1}{2L} + \frac{1}{L-1}\Big)(L')^2 - \frac{L'}{t} + \frac{(L-1)^2(AL+B/L)}{t^2} + \frac{CL}{t} + \frac{DL(L+1)}{L-1},$$

$$\text{PVI:}\quad L'' = \frac{1}{2}\Big(\frac{1}{L} + \frac{1}{L-1} + \frac{1}{L-t}\Big)(L')^2 - \Big(\frac{1}{t} + \frac{1}{t-1} + \frac{1}{L-t}\Big)L' + \frac{L(L-1)(L-t)}{t^2(t-1)^2}\Big(A + \frac{Bt}{L^2} + \frac{C(t-1)}{(L-1)^2} + \frac{Dt(t-1)}{(L-t)^2}\Big).$$

The four parameters (A, B, C, D) are fitted by an overdetermined linear system at four interior t-points and validated at the two outer points; the maximum validation residual is reported as the matrix entry. A genuine identification (HIT) requires residual $\leq 10^{-50}$.

A.5. **Per-family residual and Stokes-exponent tables.** For each of the four newly probed $\Delta < 0$ families (QL02, QL06, QL15, QL26) we report the full 2×3 Painlevé residual matrix and the 2×6 Stokes-exponent matrix in Tables 8–9. Cells in italics fall below the AMBIG threshold 10^{-20}; no cell does. The 5-pt-vs-7-pt stencil disagreement (column "stencil") is reported as the maximum across both deformations and all four derivatives.

Family	Δ	Def.	P-III	P-V	P-VI	stencil
QL02	-4	A	9.39×10^{-4}	9.64×10^{-3}	0.129	2.36×10^{-7}
QL02	-4	B	6.07×10^{-2}	4.83×10^{-2}	1.62	2.60×10^{-5}
QL06	-7	A	1.25×10^{-4}	1.07×10^{-4}	2.05×10^{-2}	$\leq 10^{-7}$
QL06	-7	B	8.82×10^{-3}	0.133	5.45	$\leq 10^{-5}$
QL15	-20	A	1.87×10^{-2}	0.230	0.450	$\leq 10^{-7}$
QL15	-20	B	0.214	0.301	7.10	$\leq 10^{-5}$
QL26	-28	A	2.40×10^{-2}	2.49×10^{-2}	36.3	$\leq 10^{-7}$
QL26	-28	B	4.79×10^{-2}	0.884	315	$\leq 10^{-5}$

TABLE 8. Painlevé residual matrices for the four newly probed $\Delta < 0$ families. No cell achieves the HIT threshold 10^{-50} or the AMBIG threshold 10^{-20}. Residuals reported are the maximum validation residuals at the two outer points $t = \pm 0.3$ after fitting at $t \in \{\pm 0.1, \pm 0.2\}$. Working precision dps = 250, depth 1500.

The QL15 family is unusual in that the D-A arc (constant-term scaling) gives $S < 1$ uniformly, while the D-B arc (root-radius scaling) gives $S > 1$ uniformly. This is the only family in the sample with this asymmetry, and it indicates that the deformation choice is not neutral: the structural Stokes signal lives on the γ-shift arc for QL15, not on the radius-scaling arc. The

Family	Def.	$S(+0.1)$	$S(-0.1)$	$S(+0.2)$	$S(-0.2)$	$S(+0.3)$	$S(-0.3)$
QL02	A	1.155	1.165	1.215	1.245	1.279	1.339
QL02	B	0.828	0.892	0.710	0.894	**0.558**	0.926
QL06	A	1.025	1.026	1.034	1.038	1.045	1.052
QL06	B	0.713	0.763	0.556	0.700	**0.364**	0.655
QL15	A	0.943	0.949	0.838	0.871	0.708	0.785
QL15	B	1.144	1.196	1.171	1.299	1.187	1.398
QL26	A	0.639	0.633	0.488	0.469	0.321	0.282
QL26	B	0.345	0.389	0.036	0.160	**−0.324**	−0.072

TABLE 9. Stokes-exponent $S(t)$ values for the four newly probed $\Delta < 0$ families. The minimum across the table per family (in bold) gives $S_{\min}$ in Table 2. $S < 1$ on at least one deformation arc in every family; QL26 reaches a negative $S(+0.3) = -0.324$ on the D-B arc, the strongest signal in the database. Note that $S(t) > 1$ entries are also diagnostically informative: a single deformation that *lifts* S above 1 does not refute Conjecture 5.1, which only requires *some* CM-respecting deformation to give $S < 1$.

reverse is true for QL02, QL06, and QL26, where the D-B arc provides the strongest signal.

A.6. **Reproducibility URLs.** The session-level computation scripts and per-family residual JSON files are archived at the SIARC research workspace under

- `sessions/2026-04-30/QL15-STOKES/` (QL15 baseline);
- `sessions/2026-05-01/STOKES-FAMILY-EXTEND-3X/` (QL02, QL06, QL26 extensions);

together with the shared engine `stokes_session_b.py` that runs the six-step Session B template given a family-spec dictionary. The $F(2,4)$ Trans closed-form database is at `sessions/2026-04-13/F24-TRANS-TABLE/`. Files are available from the author on request, in line with the AEAL reproducibility requirement. residual JSON files are archived at the SIARC research workspace under

- `sessions/2026-04-30/QL15-STOKES/` (QL15 baseline);
- `sessions/2026-05-01/STOKES-FAMILY-EXTEND-3X/` (QL02, QL06, QL26 extensions);

together with the shared engine `stokes_session_b.py` that runs the six-step Session B template given a family-spec dictionary. The $F(2,4)$ Trans closed-form database is at `sessions/2026-04-13/F24-TRANS-TABLE/`. Files are available from the author on request, in line with the AEAL reproducibility requirement.

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